

Exercise 4.1

Q7: Find the factors of

(i), (ii), (iii), (iv), (v), (vi)



(i)

$$(x^2 + 2xy + y^2) - 9z^4$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

$$(x + y)^2 - (3z^2)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(x + y - 3z^2)(x + y + 3z^2)$$

(ii)

$$(4a^2 + 8ab^2 + 4b^4) - 9c^2$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

$$(2a + 2b^2)^2 - (3c)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(2a + 2b^2 - 3c)(2a + 2b^2 + 3c)$$

(iii)

$$16d^4 - (c^4 - 2c^2d + d^2)$$

Solution:

Since, $a^2 - 2ab + b^2 = (a - b)^2$

$$(4d^2)^2 - (c^2 - d)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(4d^2 - c^2 + d)(4d^2 + c^2 - d)$$

(iv)

$$4(x^2 + 2xy^2 + y^4) - 9y^6$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

$$(2x + 2y^2)^2 - (3y^3)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(2x + 2y^2 - 3y^3)(2x + 2y^2 + 3y^3)$$

(v)

$$x^2 - y^2 - 4x - 2y + 3$$

Solution:

(Detailed solution in the end)

Since, $a^2 - 2ab + b^2 = (a - b)^2$ and $a^2 + 2ab + b^2 = (a + b)^2$

$$(x - 2)^2 - (y + 1)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(x - y - 3)(x + y - 1)$$

(vi)

$$4x^2 - y^2 - 2y - 1$$

Solution:

Since, $a^2 + 2ab + b^2 = (a + b)^2$

$$(2x)^2 - (y + 1)^2$$

Since, $a^2 - b^2 = (a - b)(a + b)$

Answer:

$$(2x - y - 1)(2x + y + 1)$$

Exercise 4.1

Q8:Factorize

(i), (ii), (iii), (iv)



(i)

$$(\sqrt{ab})^2 - (\sqrt{c})^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = \sqrt{ab}, \quad b = \sqrt{c}$$

Answer: $(\sqrt{ab} - \sqrt{c})(\sqrt{ab} + \sqrt{c})$

(ii)

$$(\sqrt{4x})^2 - (\sqrt{9y})^2$$

Solution:

$$\text{Since, } a^2 - b^2 = (a - b)(a + b)$$

$$\text{Here, } a = 2\sqrt{x}, \quad b = 3\sqrt{y}$$

Answer: $(2\sqrt{x} - 3\sqrt{y})(2\sqrt{x} + 3\sqrt{y})$

(iii)

$$(\sqrt{yz})^2 - \left(\frac{1}{\sqrt{yz}}\right)^2$$

Solution:

Since, $a^2 - b^2 = (a - b)(a + b)$

Here, $a = \sqrt{yz}$, $b = \frac{1}{\sqrt{yz}}$

Answer: $\left(\sqrt{yz} - \frac{1}{\sqrt{yz}}\right)\left(\sqrt{yz} + \frac{1}{\sqrt{yz}}\right)$

(iv)

$$(\sqrt{xzt})^2 - \left(\frac{1}{\sqrt{t}}\right)^2$$

Solution:

Since, $a^2 - b^2 = (a - b)(a + b)$

Here, $a = \sqrt{xzt}$, $b = \frac{1}{\sqrt{t}}$

Answer: $\left(\sqrt{xzt} - \frac{1}{\sqrt{t}}\right)\left(\sqrt{xzt} + \frac{1}{\sqrt{t}}\right)$